

Optimal Classification of Low-volume Medical Data with Adaptive Kernel Design

Masoud Vahid Dastgerdi
Higher School of Economics
masoud.vahid10@gmail.com

Oleg Stanislavovich Pianykh
Massachusetts General Hospital
opianykh@mgh.harvard.edu

Abstract

Accurate cancer localization in medical images is often approached with high-capacity convolutional neural networks, but their opacity and data requirements can limit their utility in settings where interpretability and constrained supervision are important. We present an interpretable kernel-learning framework for patch-level cancer classification from pixel-level annotations, with primary emphasis on a *minimal bank of learned radial kernels*. Rather than committing to one circularly symmetric filter, the method learns K radial profiles jointly, encodes each patch by three pooled response statistics per kernel, and then compresses the bank: kernels are compared by the correlation of their convolutional responses on real training patches, hierarchically clustered, and sign-aligned and merged into the smallest bank whose validation quality matches the full bank. This yields a classifier defined by a handful of explicitly plotted radial profiles rather than an opaque filter stack. We evaluate on two datasets with patient-level splits: a breast TIFF cohort of 10,070 annotation-derived patches and the LGG brain MRI cohort of 110 patients and 10,000 patches. On breast data, merging reduces 16 learned kernels to 9 with no loss, reaching 0.9415 test ROC-AUC and 0.8727 test accuracy, against 0.9212 and 0.8524 for the best single learned radial kernel; on brain MRI the 16-kernel bank improves test ROC-AUC from 0.6903 to 0.7624 over its single-kernel counterpart, with 0.7335 retained after merging to 9 kernels. A fully learned patch CNN reference reaches 0.9626 test ROC-AUC on breast data. We further show that much of the gain over a single kernel comes from the multi-statistic readout rather than from kernel count alone, which clarifies where the capacity of such compact models actually resides.

1 Introduction

Medical image classification systems are increasingly used to support cancer diagnosis, lesion triage, and downstream localization. Deep convolutional models can achieve strong predictive performance, but they typically require substantial data and their learned internal representations are difficult to interpret directly. This tension is especially relevant for clinical workflows in which models should not only provide accurate decisions but also expose evidence

that can be inspected by researchers or clinicians.

This work studies a complementary approach: instead of learning all visual primitives end-to-end, we construct a large bank of interpretable convolutional kernels and learn a classifier over their patch-level responses. The central hypothesis is that a diverse bank of biologically and radiologically meaningful texture, edge, blob, and local-contrast filters can capture discriminative differences between lesion and non-lesion tissue while preserving explicit filter semantics.

The repository implements an end-to-end patch-level cancer classification pipeline. Pixel-level red-contour annotations are converted to binary malignant masks; positive patches are sampled from lesion regions and negative patches are sampled from background regions. Each patch is convolved with kernels from multiple analytic families, and each kernel response is summarized by a scalar activation statistic. Candidate kernels are ranked using ROC-AUC and Fisher score, then selected with maximal marginal relevance (MMR) to reduce redundancy. A logistic or shallow multilayer perceptron classifier is trained on the selected kernel-response features.

The main contributions are:

- We formulate an interpretable learned radial-kernel framework for cancer patch classification in which the primary model is a *minimal bank* of jointly learned circularly symmetric profiles rather than a single filter.
- We introduce a redundancy-merging procedure that measures kernel similarity by the correlation of convolutional responses on real training patches, clusters kernels hierarchically, merges each cluster into one sign-aligned radial profile, and selects the smallest bank whose validation quality stays within tolerance of the full bank.
- We show that a multi-statistic readout (mean absolute, absolute maximum, and signed top- k response per kernel) accounts for a large share of the improvement over single-kernel models, and we separate this readout effect from the effect of kernel count.
- We implement a reproducible patient-level experimental pipeline with annotation-derived patch extraction, train/validation/test splits, feature caching, model checkpoints, ROC/PR curves, confusion matrices, and heatmap visualization.

- We validate the method on two modalities with different acquisition physics, a breast TIFF cohort and the LGG brain MRI cohort, and report where the procedure transfers and where it does not.
- We empirically compare the minimal radial bank with single learned radial kernels, compact selected-kernel models, larger selected-kernel banks, abess-based sparse selection, composite kernels, triangular/square shape-sensitive kernels, and a fully learned patch CNN baseline.
- We incorporate report-level analyses of Sobol QMC kernel sampling, radial-kernel variants, and rotation-invariant kernel averaging.

2 Related Work

Classical medical image analysis has long used engineered features, including edges, blobs, texture descriptors, and co-occurrence statistics, to characterize tissue appearance [1, 2]. Gaussian derivatives, Laplacian-of-Gaussian filters, and difference-of-Gaussians filters are standard tools for multiscale smoothing, edge detection, and blob detection [3]. Gabor filters represent oriented frequency-selective texture patterns and have been widely used in biomedical imaging [4]. Histogram-of-oriented-gradients descriptors capture local edge-orientation distributions [5], while local binary patterns and gray-level co-occurrence matrices model local texture and spatial intensity relationships [6, 7].

Several cancer-classification studies have shown that handcrafted texture representations can remain useful when training sets are small. Doyle et al. used Gabor and architectural descriptors with an SVM for breast histopathology grading [13]. Ojansivu et al. combined LBP/LPQ texture descriptors with kernel SVMs for breast cancer morphology classification [14]. Gopinath and Shanthi reported a small-data thyroid cytology pipeline based on Gabor statistical features and SVM classification [15]. Related breast and cervical cancer studies have also used Gabor, co-occurrence, morphological, or other handcrafted descriptors with classical classifiers [16, 17, 18]. These works motivate the present focus on explicit filter responses, but they generally rely on fixed descriptor families rather than systematic generation, ranking, and diversity-aware selection of a broad analytic kernel bank.

Deep convolutional neural networks learn such filters automatically and have become dominant in medical image classification [8, 9]. However, end-to-end learning often trades interpretability for performance, and learned filters may be difficult to map to known visual structures. Hybrid approaches that combine fixed or constrained feature banks with learned classifiers provide an intermediate point in this design space: the low-level representation remains inspectable, while the classifier can still learn task-specific decision boundaries. Recent examples include fusing deep features with texture descriptors for renal can-

cer detection [19] and feed-forward breast histopathology models that use logistic-regression or SVM modules rather than standard back-propagation through a large CNN [20].

Feature selection is another relevant line of work. Sparse model selection and L0-regularized methods aim to identify compact predictive subsets [10]. Maximal marginal relevance, originally introduced for information retrieval, selects items by balancing relevance and diversity [11]. In the present setting, relevance corresponds to cancer-versus-background discriminative strength, while diversity corresponds to low redundancy among kernel response vectors.

3 Methodology

3.1 Problem Formulation

Let an image patch be denoted by $x \in \mathbb{R}^{H \times W}$, with binary label $y \in \{0, 1\}$, where $y = 1$ indicates a patch sampled from a malignant annotation mask and $y = 0$ indicates background or normal tissue. Given a training set $\mathcal{D} = \{(x_i, y_i)\}_{i=1}^N$, the goal is to learn a classifier $f(x)$ that estimates $p(y = 1 | x)$.

Instead of feeding raw pixels directly into a deep model, we define a bank of analytic kernels

$$\mathcal{K} = \{k_j\}_{j=1}^M, \quad k_j \in \mathbb{R}^{s \times s},$$

and map each patch to a response vector

$$\phi(x) = [r_1(x), r_2(x), \dots, r_K(x)] \in \mathbb{R}^K,$$

where $K \leq M$ is the number of selected kernels. The default response statistic is the maximum absolute convolutional activation:

$$r_j(x) = \max_{u,v} |(x * k_j)_{u,v}|.$$

The repository also supports signed maximum and mean absolute response functions.

3.2 Patch Extraction from Annotations

Annotation images contain red malignant contours. The implementation thresholds the red channel while suppressing green and blue channels, applies morphological dilation, erosion, closing, and hole filling, and obtains a binary lesion mask. Positive patches are sampled from regions whose mask coverage exceeds a configured threshold. Negative patches are sampled outside the lesion, with support for near-lesion and far-lesion sampling. In the main experiments, the patch size is 128 x 128 pixels, positives require at least 0.8 lesion-mask coverage, negatives allow at most 0.02 lesion-mask coverage, and patches with low nonzero intensity fraction are filtered.

Patient or study identifiers are inferred from filenames, and splits are assigned at the image/patient level using a 70/15/15 train/validation/test ratio. This prevents patches from the same source image from appearing simultaneously in training and held-out evaluation splits.

3.3 Kernel Families

The main kernel bank contains nine analytic families: Gaussian filters for local smoothing; anisotropic Gaussian filters for elongated structures; difference-of-Gaussians filters for multiscale contrast; Laplacian-of-Gaussian filters for blob-like structures; Gabor filters for oriented texture; HOG-inspired derivative filters for local gradient orientation; GLCM-inspired two-point offset filters for spatial co-occurrence; LBP-inspired center-neighbor contrast filters; and MRF-inspired discrete Laplacian filters for local smoothness.

Each family is sampled over a family-specific parameter space. The implementation supports random sampling, Sobol quasi-Monte Carlo sampling, and Latin hypercube sampling. Unless otherwise noted, experiments use kernels of size 31×31 .

3.4 Learned Radial Kernels

The compact models in this paper constrain every convolutional filter to be circularly symmetric and learn a radial profile $f(r)$. Instead of learning an unconstrained $s \times s$ kernel, all pixels at the same distance from the kernel center share a parameter. Writing $B_r \in \{0, 1\}^{s \times s}$ for the indicator of radial shell r , a kernel is recovered as

$$k = \sum_{r=0}^R f(r) B_r,$$

so an $s \times s$ filter is described by $R + 1$ numbers. This reduces the degrees of freedom, makes the learned structure directly plottable, and enforces an isotropic inductive bias appropriate for blob-like and halo-like tissue patterns. Shells beyond a configured radius $R_{\max}$ are frozen at their seed values, which bounds the effective support.

The single-kernel variant learns one such profile. Additional signed compact variants use a signed response, hard radial cutoff, monotonicity/decay regularization, and scale normalization to improve interpretability at some cost in AUC.

3.5 Radial Kernel Bank

The primary model generalizes this to K profiles $\{f_1, \dots, f_K\}$ learned jointly, producing a bank $\{k_1, \dots, k_K\}$ that remains radial by construction. Profiles are seeded from two sources to span different spatial scales: variants of the rotation-invariant profile extracted from the composite kernel, and synthetic center-surround shapes of increasing width. Training minimizes a class-balanced binary cross-entropy through a small multilayer perceptron head, regularized by

$$\mathcal{L} = \mathcal{L}_{\text{cls}} + \lambda_s \mathcal{L}_{\text{smooth}} + \lambda_c \mathcal{L}_{\text{curv}} + \lambda_a \mathcal{L}_{\text{anchor}} + \lambda_n \mathcal{L}_{\text{scale}} + \lambda_d \mathcal{L}_{\text{div}},$$

where the smoothness and curvature terms penalize first and second differences of f along r , the anchor term limits drift from the seed, the scale term keeps profile norms comparable, and the diversity term $\mathcal{L}_{\text{div}} = \text{mean exp}(-\|f_i -$

$f_j\|^2)$ discourages profiles from collapsing onto one another. The checkpoint is selected by validation ROC-AUC, after which a class-balanced logistic model is refit on the frozen bank to provide calibrated probabilities.

3.6 Multi-Statistic Response Encoding

A single scalar per kernel discards most of the response map. Each kernel therefore contributes three pooled statistics of its response $z = x * k_j$:

$$\text{mean}|z|, \quad \max|z|, \quad \frac{1}{m} \sum_{\text{top-}m} z - \frac{1}{m} \sum_{\text{top-}m} (-z),$$

with m a small fraction of the response map. The third statistic is signed, so it distinguishes bright-on-dark from dark-on-bright structure that the absolute statistics conflate. A bank of K kernels thus yields $3K$ features. Section 5.2 shows this readout is responsible for a substantial part of the gain over single-kernel models.

3.7 Redundancy Merging to a Minimal Kernel Set

Jointly trained profiles frequently converge to near-duplicates, or to sign-flipped copies that carry identical information after pooling. We compress the bank post hoc in four steps.

Similarity in response space. For each kernel we collect its convolutional response over a fixed subsample of training patches and compute the Pearson correlation ρ_{ij} between kernels. Redundancy is measured as $|\rho_{ij}|$, so a kernel and its sign-flipped twin count as identical. Response correlation is used rather than kernel-space cosine similarity because the classifier only ever observes responses, and because natural image statistics make spatially distinct kernels produce correlated responses; orthogonality in kernel space would not imply decorrelation on real tissue.

Clustering. Kernels are clustered by average-linkage hierarchical clustering on the distance $1 - |\rho_{ij}|$.

Merging. Each cluster is collapsed to one profile. Members are sign-aligned to the cluster medoid, averaged, and renormalized to the members' mean norm. Because the merge is performed on the radial coefficients and re-expanded through the shell basis, the merged filter is still exactly radial.

Model selection. We sweep every achievable cluster count $K' = K, \dots, 1$, refit the same class-balanced logistic head on each reduced bank, and select the *smallest* K' whose validation ROC-AUC and validation threshold-tuned accuracy both remain within a tolerance τ of the full bank. All selection is performed on validation data; test data is touched only for final reporting.

3.8 Kernel Ranking and Selection

For each candidate kernel, the method computes response distributions for positive and negative patches. Two uni-

variate scores are used:

$$\text{AUC}(k_j) = \text{ROC-AUC}(\{r_j(x_i)\}, \{y_i\}),$$

and the Fisher score

$$F(k_j) = \frac{(\mu_j^+ - \mu_j^-)^2}{(\sigma_j^+)^2 + (\sigma_j^-)^2 + \epsilon}.$$

The top M' candidates are ranked by AUC and Fisher score. MMR then selects a subset iteratively:

$$k^* = \arg \max_{k_j \notin S} \lambda \text{AUC}(k_j) - (1 - \lambda) \max_{k_\ell \in S} |\rho(r_j, r_\ell)|,$$

where S is the selected set, ρ is Pearson correlation between response vectors, and $\lambda = 0.75$ in the main experiments.

3.9 Classifier

The selected response vector $\phi(x)$ is passed to either a logistic classifier or a shallow MLP. The primary reported kernel-bank experiments use a logistic classifier trained with binary cross-entropy using Adam. Feature standardization is enabled in several later experiments. The CNN baseline uses a four-block convolutional network with batch normalization, ReLU, max pooling, dropout, adaptive average pooling, and a two-layer linear head trained with cross-entropy and AdamW.

3.10 Algorithm

1. Convert annotation contours into binary lesion masks.
2. Sample positive patches from lesion regions and negative patches from background/normal tissue.
3. Split images at patient/study level into train, validation, and test sets.
4. Generate analytic kernel bank from the chosen families.
5. For every kernel, compute scalar responses on all training patches.
6. Rank kernels by ROC-AUC and Fisher score.
7. Select kernels with MMR to balance discrimination and response diversity.
8. Train a logistic or shallow neural classifier on selected responses.
9. Evaluate ROC-AUC, accuracy, precision, recall, F1, and confusion matrices on held-out splits.

For the minimal radial bank, steps 4–7 are replaced by direct optimization of the bank:

1. Seed K radial profiles from rotation-invariant and synthetic center-surround shapes.
2. Jointly learn all profiles with a class-balanced objective and radial regularization; select the checkpoint by validation ROC-AUC.

Table 1: Patch counts for both cohorts. Splits are assigned at the patient/image level, so no source image contributes to more than one split.

Cohort	Split	Pos.	Neg.	Total
Breast	Train	2,900	3,430	6,330
	Validation	1,080	790	1,870
	Test	1,040	830	1,870
	Total	5,020	5,050	10,070
Brain LGG	Train	3,000	3,000	6,000
	Validation	1,000	1,000	2,000
	Test	1,000	1,000	2,000
	Total	5,000	5,000	10,000

3. Encode each patch by three pooled statistics per kernel and refit a class-balanced logistic head.
4. Compute pairwise response correlations, cluster on $1 - |\rho|$, and merge clusters into sign-aligned radial profiles.
5. Sweep $K' = K, \dots, 1$ and retain the smallest bank within tolerance τ of full-bank validation quality.

4 Experimental Setup

4.1 Datasets

We evaluate on two cohorts that differ in modality, acquisition physics, and annotation style. This separation matters because it tests whether the merging procedure is a property of the method or of one dataset.

Breast TIFF cohort. The primary cohort consists of high-resolution TIFF images with expert red-contour malignant annotations, comprising 127 malignant images, 365 normal images, and 125 annotation files. Images are approximately 6000×4750 pixels. Annotation contours are converted to binary lesion masks, and 128×128 patches are extracted at native resolution, giving 10,070 patches in total. Positive patches require at least 0.8 lesion coverage and negatives at most 0.02, with both near-lesion and far-lesion negatives sampled.

Brain LGG MRI cohort. The second cohort is the lower-grade glioma MRI collection of Buda et al. [21], derived from TCGA/TCIA, containing 110 patients with per-slice FLAIR images and manually delineated tumor masks stored as paired `.tif` files. Patches of 64×64 pixels are sampled directly from the provided masks, with positives requiring at least 0.80 tumor coverage and negatives strictly zero coverage; a fraction of negatives is drawn from a dilated peritumoral band so that the negative class contains hard near-boundary examples. A minimum tissue-occupancy constraint suppresses background-air patches. This yields a balanced set of 10,000 patches.

Both cohorts use a 70/15/15 patient-level split with a fixed seed. The slight imbalance in the breast splits follows from patient-level assignment and patch-extraction success

rates; the brain cohort is balanced by construction through per-class caps. Patches are loaded in grayscale.

An important preprocessing difference between the two cohorts is per-patch intensity standardization. It is applied for brain MRI, where scanner-dependent intensity offsets carry no diagnostic meaning, and omitted for the breast cohort, where absolute intensity is informative. Section 5.2 quantifies the cost of getting this choice wrong.

4.2 Evaluation Metrics

The primary metric is ROC-AUC because it is threshold-independent and robust to moderate class imbalance. Accuracy is reported for all kernel-bank experiments. For the CNN baseline, precision, recall, F1, ROC-AUC, accuracy, loss, and confusion matrices are available.

4.3 Baselines and Variants

We evaluate MMR-selected analytic kernel banks, abess sparse selection, composite kernels, triangular/square shape-sensitive kernels, learned radial kernels, rotation-invariant averaged kernels, and a learned patch CNN baseline.

4.4 Implementation Details

The principal MMR configurations use nine kernel families, 31×31 kernels, an absolute-maximum response statistic, 60 training epochs, batch size 64, learning rate 5×10^{-4} , and patient-level train/validation/test evaluation. The MMR parameter is $\lambda = 0.75$. The strongest reported interpretable configurations use larger selected banks, such as 90 selected kernels from a broader candidate pool. Compact models with $K = 1$, $K = 3$, and $K = 5$ are also included.

The CNN baseline uses 128×128 RGB inputs, batch size 64, AdamW with learning rate 3×10^{-4} , weight decay 10^{-4} , patience 6, and a maximum of 25 epochs. The recorded best model occurs by epoch 6 based on validation ROC-AUC. The March and April reports include additional single-radial-kernel experiments; these are treated as separate ablations because they optimize kernel parameters directly rather than selecting from the fixed multi-kernel bank.

For the minimal radial bank, both cohorts use $K = 16$ initial kernels, three pooled statistics per kernel, a 32-unit hidden head with dropout 0.1, batch size 128, Adam, and checkpoint selection by validation ROC-AUC. The breast configuration uses 31×31 kernels with radial shells learned to $R_{\max} = 20$, 25 epochs, and profile learning rate 2×10^{-3} chosen on validation ROC-AUC from $\{10^{-2}, 2 \times 10^{-3}, 5 \times 10^{-4}\}$; patches are 128×128 and are not intensity-standardized. The brain configuration uses 63×63 kernels with $R_{\max} = 30$, 40 epochs, and 64×64 standardized patches. Merging uses a subsample of training patches for the response-correlation estimate and a tolerance of $\tau = 0.005$ on both validation ROC-AUC and val-

idation threshold-tuned accuracy. Decision thresholds are tuned on validation only.

5 Results

5.1 Main Quantitative Results

Table 2 summarizes representative held-out results on the breast cohort. The table is organized around the minimal merged radial bank, which is the central model of this paper, followed by the CNN reference and the remaining interpretable baselines sorted by test ROC-AUC. The minimal bank uses nine radial profiles and reaches 0.9415 test ROC-AUC and 0.8727 test accuracy, exceeding both the best single learned radial kernel and every MMR-selected analytic bank, including configurations that use ten times as many kernels. The CNN baseline remains the strongest absolute performer with 0.9626 test ROC-AUC and 0.9021 test accuracy, but the gap to the minimal bank is now 0.021 ROC-AUC, against a model defined by nine plottable one-dimensional profiles.

These results support four observations. First, a small bank of learned radial profiles outperforms both single radial kernels and much larger MMR-selected analytic banks, so kernel count is a poor proxy for representational quality. Second, learned CNN features still provide the best absolute accuracy and AUC in the available experiments. Third, the gap between the minimal bank and the CNN is modest in ROC-AUC, suggesting that explicit radial texture and morphology cues capture a substantial fraction of the discriminative signal. Fourth, the validation-tuned accuracy of the merged bank exceeds that of every analytic bank in the table, indicating that the calibration instability seen in compact MMR models is largely resolved by the class-balanced logistic head.

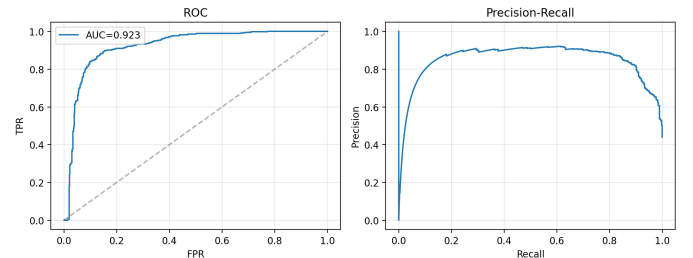


Figure 1: Held-out ROC and precision-recall curves for the selected-kernel classifier.

Table 2: Held-out performance of representative models. The primary model of interest is bolded; the strongest reference and selected-bank results are also emphasized.

Model / configuration	Features	Val AUC	Val Acc.	Test AUC	Test Acc.
Minimal merged radial bank (primary focus)	9	0.9373	0.8856	0.9415	0.8727
Radial kernel bank before merging	16	0.9388	0.8861	0.9411	0.8754
Learned single radial kernel	1	0.9189	0.8545	0.9286	0.8668
Patch CNN reference	learned	0.9445	0.8818	0.9626	0.9021
Selected kernel bank with QMC sampling	90	0.9270	0.8603	0.9387	0.8866
Selected kernel bank, variant A	90	0.9271	0.8636	0.9377	0.8882
Selected kernel bank, variant B	90	0.9273	0.8620	0.9376	0.8888
Larger selected kernel bank	200	0.9159	0.8514	0.9313	0.8578
Sparse ABESS-selected bank	200	0.9095	0.7849	0.9236	0.7690
Shape-sensitive triangular/square bank	24	0.9109	0.8481	0.9221	0.8465

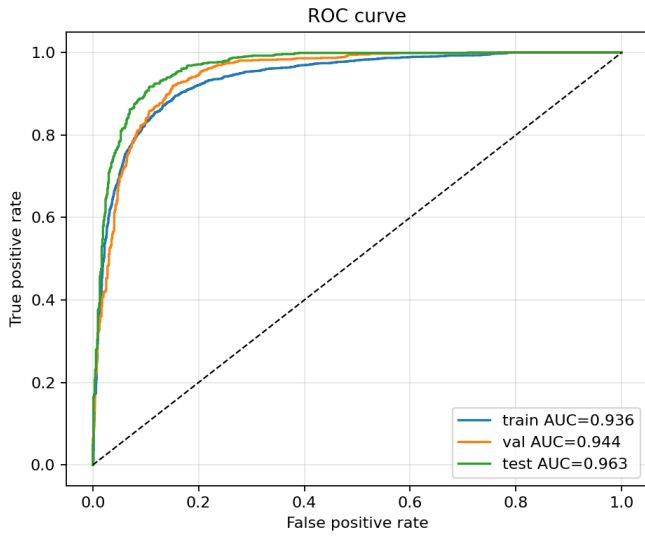


Figure 2: Held-out ROC curve for the patch CNN reference model.

5.2 Minimal Radial Kernel Banks

Table 3 reports the central result of this paper. On both cohorts, jointly learning a bank of radial profiles clearly outperforms the corresponding single learned radial kernel, and the merging procedure of Section 3.7 then removes roughly half the kernels essentially for free.

The single-kernel reference in Table 3 is the strongest saved single radial run under the same experiment directory and radial support $R_{\max} = 20$ used for the bank, namely the monotone-decreasing variant at 0.9212 test ROC-AUC. Across all radial supports the best single-kernel variant we can reproduce from saved artifacts reaches 0.9227, so the reported improvement is insensitive to this choice.

On the breast cohort the 16-kernel bank raises test ROC-AUC from 0.9212 to 0.9411 and test accuracy from 0.8412 to 0.8749 relative to the strongest single radial kernel. Merging to 9 kernels, a 44% reduction, leaves test ROC-AUC statistically unchanged at 0.9415 and test accuracy

at 0.8733. One merged cluster absorbed six redundant profiles while five kernels remained singletons, indicating that the trained bank contained a small number of genuinely distinct radial shapes surrounded by near-duplicates.

On the brain cohort the same procedure improves test ROC-AUC from 0.6903 to 0.7624, a larger relative gain than on breast data, confirming that the benefit of the bank is not specific to one modality. Merging behaves less favorably here: the validation-selected bank of 9 kernels retains 0.7801 validation ROC-AUC but drops to 0.7335 on test, a loss of 0.029 that the validation tolerance did not predict. We report this asymmetry rather than tuning it away. It indicates that with a noisier decision boundary the validation split is too small to certify a merge, and that the tolerance criterion should be tightened or replaced by cross-validated selection when the validation AUC gap between candidate banks is comparable to its own sampling error.

5.2.1 Where the Capacity Actually Resides

Because the merge sweep evaluates every K' , it doubles as an ablation on kernel count with the readout held fixed. Table 4 reports representative points.

The breast column is the informative one. A bank merged all the way down to a *single* radial kernel still reaches 0.9346 test ROC-AUC, above the 0.9212 obtained by the best directly optimized single radial kernel. The two models have the same number of filters; they differ only in that the merged model reads three pooled statistics from its kernel and applies a class-balanced logistic head, whereas the baseline reads one absolute-maximum scalar. This isolates the readout as the source of roughly two thirds of the total improvement, with the remaining gain from 1 to 9 kernels contributing about 0.007 ROC-AUC and 0.025 accuracy.

This finding is reinforced by the optimization traces: on the breast cohort the validation-selected checkpoint occurs at the first epoch under every profile learning rate we tried, meaning the seeded profiles were already close to their useful configuration and further profile learning did not improve validation AUC. The practical implication is that

Table 3: Minimal radial kernel banks against single learned radial kernels. “Feat.” is the number of scalar features seen by the classifier. The merged banks are selected on validation data alone, using tolerance $\tau = 0.005$. Test accuracy is reported at the validation-tuned threshold.

Cohort	Model	Kernels	Feat.	Val AUC	Test AUC	Test Acc.
Breast	Best single learned radial kernel	1	1	0.9100	0.9212	0.8412
	Radial kernel bank	16	48	0.9388	0.9411	0.8749
	Minimal merged bank	9	27	0.9373	0.9415	0.8733
Brain LGG	Single learned radial kernel	1	3	0.7281	0.6903	0.6365
	Radial kernel bank	16	48	0.7849	0.7624	0.6750
	Minimal merged bank	9	27	0.7801	0.7335	0.6640

Table 4: Merge sweep on both cohorts. Each row refits the same class-balanced logistic head on a bank merged to K' kernels. Rows marked * are the validation-selected operating points.

K'	Breast		Brain LGG	
	Val AUC	Test AUC	Val AUC	Test AUC
16	0.9389	0.9411	0.7850	0.7624
13	0.9388	0.9412	0.7740	0.7593
11	0.9364	0.9407	0.7755	0.7531
9*	0.9373	0.9415	0.7801	0.7335
8	0.9365	0.9405	0.7794	0.7434
6	0.9299	0.9356	0.7593	0.7256
4	0.9295	0.9349	0.7220	0.7098
2	0.9259	0.9326	0.7039	0.6964
1	0.9271	0.9346	0.6218	0.6671

for compact radial models, enriching the per-kernel read-out is a cheaper and more reliable lever than adding kernels or training profiles longer. The brain column shows the opposite balance, with $K' = 1$ collapsing to 0.6671 test ROC-AUC; there, kernel count genuinely matters, which is consistent with tumor appearance on FLAIR varying over a wider range of spatial scales than the breast lesions.

5.2.2 Effect of Patch Standardization

Per-patch intensity standardization is required on brain MRI but harmful on breast data. Under an otherwise identical breast configuration, enabling standardization reduces test ROC-AUC from 0.9411 to 0.8605 and test accuracy from 0.8754 to 0.7898. Standardization removes the per-patch mean and scale, which discards absolute-intensity contrast that is diagnostically meaningful in the breast cohort while being an acquisition artifact in MRI. This is a substantial effect, larger than the gap between any two model variants in Table 3, and it argues against treating preprocessing as a modality-independent default.

5.3 Compact Kernel Ablations

Table 5 reports compact MMR models. Even a single selected kernel reaches 0.9157 test ROC-AUC, and the best three-kernel configuration reaches 0.9239 test ROC-AUC.

This is notable because the classifier operates on only one to three scalar responses per patch.

The compact runs show that individual kernels can be strongly discriminative, but thresholded accuracy varies substantially. This indicates that the raw ranking objective is aligned with AUC but not necessarily with calibration or the default 0.5 decision threshold. The learned radial-kernel reports reinforce this point. A directly optimized unconstrained radial kernel reaches 0.9286 test AUC and 0.8668 test accuracy, outperforming several compact selected-kernel variants. A more constrained signed compact radial kernel improves interpretability by enforcing signed response, limited support, and smoother radial decay, but its reported test AUC drops to 0.8808. An independently initialized smooth signed radial model reports 0.9187 test AUC and 0.848 test accuracy. These results suggest a performance-interpretability tradeoff inside the single-kernel design space.

5.4 Sampling Strategy: Random versus QMC

The QMC report compares a random-sampling baseline against Sobol low-discrepancy sampling under matched high-capacity MMR/composite settings: 4,500 total kernels, $K = 90$ selected base kernels, and eight composite kernels. QMC improves all reported test metrics in that comparison.

This supports Sobol sampling as a useful design choice for covering heterogeneous kernel-parameter spaces more evenly than independent random sampling.

5.5 Composite Kernel Analysis

Composite kernels are constructed by grouping selected kernels and summing them with normalization. In the available experiments, composite models are consistently weaker than their non-composite counterparts. For example, one 90-kernel selected-bank model obtains 0.9376 test AUC, whereas its eight-composite representation obtains 0.9026 test AUC. The strongest 90-kernel selected bank obtains 0.9387 test AUC before composition and 0.8934 after composition.

This suggests that naive weighted summation destroys

Table 5: Compact-kernel MMR ablations.

Compact configuration	Kernels	Val AUC	Val Acc.	Test AUC	Test Acc.
Single-kernel selected model	1	0.8994	0.8307	0.9157	0.8449
Single-kernel model from a larger candidate short-list	1	0.9010	0.8330	0.9179	0.8428
Three-kernel selected model with stronger validation accuracy	3	0.9085	0.8497	0.9101	0.8417
Three-kernel selected model with stronger test AUC	3	0.9085	0.8011	0.9239	0.7914
Five-kernel selected model	5	0.9071	0.7765	0.9234	0.7594

Table 6: Random versus Sobol QMC kernel sampling.

Evaluation	Random	QMC	Gain
Base model test AUC	0.9195	0.9387	+0.0192
Base model test accuracy	0.8646	0.8866	+0.0221
Composite test AUC	0.8855	0.8934	+0.0080
Composite test accuracy	0.8063	0.8176	+0.0114

useful complementary response information. Composite filters may still be useful for visualization or deployment simplicity, but the current uniform grouping strategy is not a substitute for retaining multiple selected response channels.

5.6 Rotation-Invariant Kernels

The rotation-invariance reports construct circularly symmetric kernels by averaging rotated versions of a composite kernel. On a newly generated patch split, a right-angle averaged invariant kernel obtains 0.8755 test AUC and 0.8006 test accuracy, compared with 0.8933 AUC and 0.8150 accuracy for the original non-invariant composite kernel. The invariant kernel gives identical metrics under 0, 90, 180, and 270 degree rotations, confirming exact right-angle stability. Averaging over every 10 degrees produces slightly lower performance in the reported run. Thus, rotation averaging offers predictable invariance but introduces a measurable performance cost.

5.7 Comparison with abess Selection

ABESS-based sparse selection provides competitive AUC but lower thresholded accuracy in several held-out runs. For example, the 200-feature ABESS-selected model reaches 0.9236 test ROC-AUC, but its test accuracy is 0.7690. Compact ABESS models with three to five selected features reach approximately 0.9216 to 0.9225 test AUC but lower accuracy. These results suggest that sparse selection finds informative features, but the downstream model may require calibration, regularization adjustment, or threshold tuning to improve operating-point performance.

5.8 Qualitative Outputs

The repository includes ROC/PR curves, confusion matrices, kernel visualizations, scatter plots, and heatmaps for multiple held-out examples. These qualitative outputs are useful for inspecting whether high-response regions align with lesion-like regions. The current repository does not include expert qualitative scoring of heatmap localization, so we treat these visualizations as diagnostic rather than as formal localization metrics.

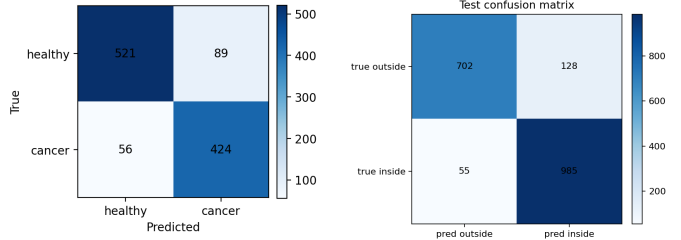


Figure 3: Confusion matrices for the selected-kernel classifier (left) and the patch CNN baseline (right).

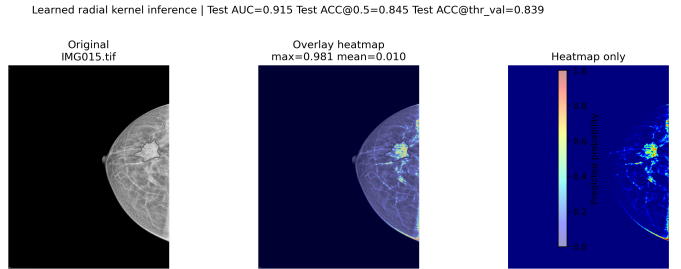


Figure 4: Example heatmap overlay for the learned single radial kernel on a held-out test image.

Table 7: Composite kernel results.

Configuration	Base kernels	Comp.	Base AUC	Comp. AUC	Comp. Acc.
Selected-bank variant B	90	8	0.9376	0.9026	0.8299
QMC selected-bank model	90	8	0.9387	0.8934	0.8176
Single-composite compression model	90	1	0.9377	0.8635	0.7797

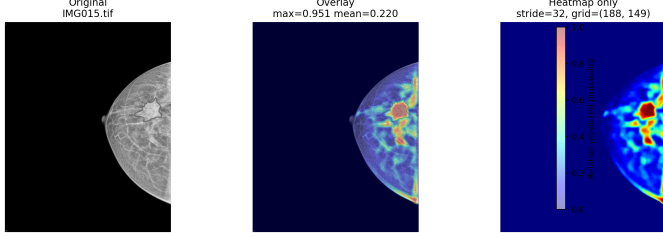
Reduced radial-kernel bank inference ($K'=9$, from 16) | Test AUC=0.942 Test ACC@0.5=0.873 Test ACC@thr=0.873

Figure 5: Full-image inference for the minimal merged radial bank ($K' = 9$) on a held-out breast image. Because the bank features are patch-level pooled statistics rather than a single dense convolution, the map is produced by scoring a sliding patch grid with exactly the classifier used at training time. This is the primary qualitative visualization for the proposed model.

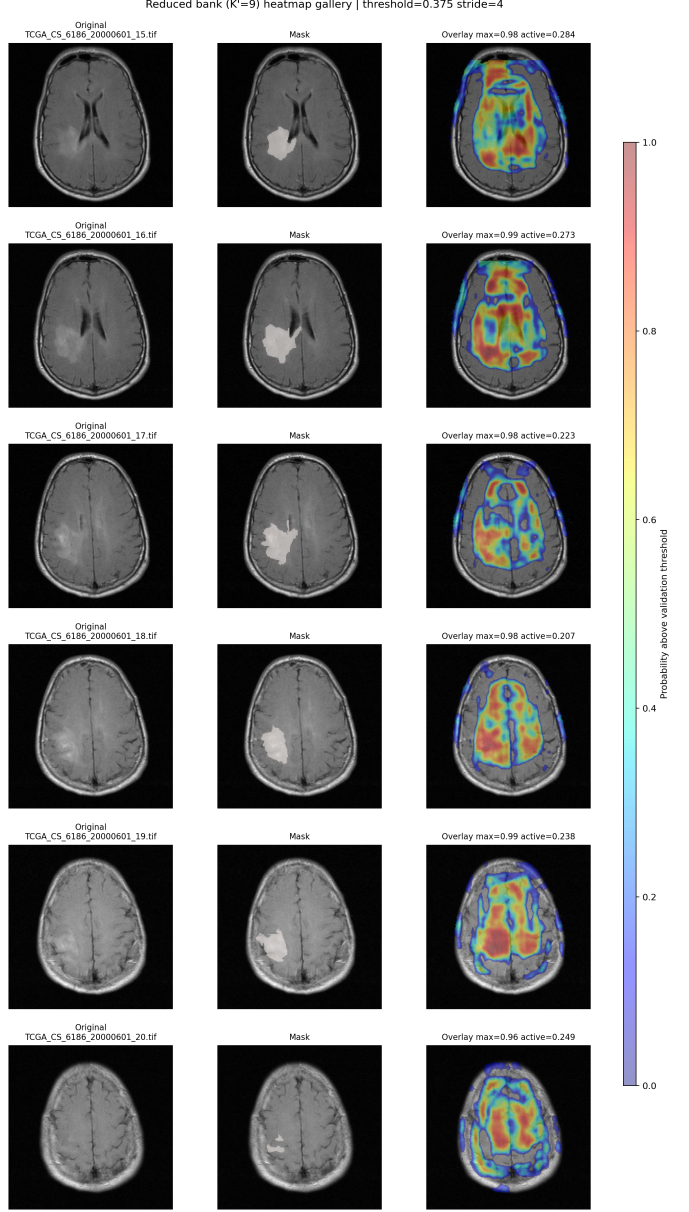


Figure 6: Minimal merged radial bank applied to held-out brain LGG MRI slices. Columns show the original slice, the ground-truth tumor mask, and the patch-center probability overlay above the validation-tuned threshold.

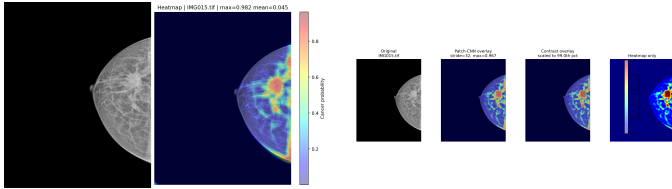


Figure 7: Additional heatmap overlays for the selected multi-kernel bank (left) and patch CNN (right) on a held-out test image. These visualizations are diagnostic only; no pixel-level localization metric is claimed.

6 Discussion

The empirical results show that a small bank of jointly learned radial profiles provides strong patch-level cancer discrimination while remaining inspectable. The merged nine-kernel model is defined by nine one-dimensional curves that can be plotted in a single figure, yet it outperforms analytic banks with ninety selected kernels. Compactness and interpretability are therefore not purchased at the cost of accuracy in this setting; the binding constraint is how much information is extracted per kernel, not how many kernels are retained.

The ablation in Section 5.2 is the most consequential result for practitioners, and it complicates the natural reading of the headline numbers. A bank merged to one kernel still exceeds the directly optimized single radial kernel, which means the improvement over prior single-kernel results is driven substantially by the three-statistic readout and class-balanced calibration rather than by having many filters. We state this explicitly because the opposite conclusion, that more kernels are the source of the gain, would be the natural inference from Table 3 alone and would misdirect follow-up work. On brain MRI the balance is different and kernel count does matter, so the readout effect should not be over-generalized either.

Merging behaved asymmetrically across the two cohorts. On breast data the validation tolerance selected a bank that also held up on test, whereas on brain MRI the selected bank lost 0.029 test ROC-AUC despite satisfying the validation criterion. The tolerance is applied to a single validation split, so when candidate banks differ by less than the sampling error of that split the criterion cannot reliably discriminate among them. Repeated-split or cross-validated selection would make the procedure more trustworthy at essentially no cost in the final model.

The compact-kernel experiments are especially informative. A one-kernel model achieving greater than 0.91 test AUC implies that certain simple local structures are strongly associated with annotated tumor patches in the current dataset. However, the lower and more variable accuracy of compact models shows that AUC alone is insufficient for deployment. Larger selected banks improve accuracy and provide more stable operating points.

The composite-kernel experiments reveal a limitation of simple feature compression. Summing selected filters into composite kernels reduces dimensionality but loses

the identity of individual response channels. Since different filters may activate on distinct lesion characteristics, collapsing them before classification can remove discriminative structure that a classifier would otherwise combine nonlinearly or with separate weights.

The CNN baseline achieves the best performance. This is expected because it learns hierarchical features directly from RGB image patches and can adapt all convolutional filters jointly. However, the CNN is less transparent: its internal filters and high-level activations are harder to interpret than the analytic kernel families. The proposed kernel-bank model therefore occupies a practical middle ground between classical handcrafted features and fully learned deep representations.

6.1 Limitations

Several limitations should be noted. First, the experiments are patch-level rather than patient-level diagnostic evaluations. Patch classification performance does not directly imply whole-image or patient-level clinical utility. Second, patch labels are derived from annotation-mask coverage and may include boundary ambiguity. Third, the results are based on available experiment artifacts rather than a single locked benchmark script with repeated seeds and confidence intervals; no confidence intervals accompany the reported differences, and the smaller gaps in Table 4 should not be treated as significant individually. Fourth, qualitative heatmap outputs are not accompanied by quantitative localization metrics such as Dice, IoU, or free-response ROC. Fifth, merge selection relies on a single validation split, which Section 5.2 shows can fail to transfer on the noisier cohort. Sixth, the CNN reference was evaluated only on the breast cohort, so the interpretability-versus-accuracy gap is not established for brain MRI. Finally, although two cohorts are now used, both are public or repository-local and neither constitutes prospective external validation on an independent clinical population.

6.2 Failure Cases

Likely failure modes include tumor patches with weak texture contrast, negative patches near lesion boundaries, normal tissue with tumor-like local texture, and images with annotation or staining variability. Compact models may also fail when a single discriminative texture cue is absent, motivating larger selected banks or hybrid models.

7 Conclusion

We presented an interpretable learned radial-kernel framework for patch-level cancer classification from annotated medical images, in which the operating model is a minimal bank of circularly symmetric profiles rather than a single filter. The method extracts annotation-derived patches, jointly learns a bank of radial profiles, encodes each patch by three pooled response statistics per kernel, and then merges redundant kernels in response-correlation

space down to the smallest bank that preserves validation quality.

On the breast cohort the procedure reduces 16 learned kernels to 9 with no measurable loss, reaching 0.9415 test ROC-AUC and 0.8727 test accuracy, compared with 0.9212 and 0.8412 for the best single learned radial kernel and 0.9626 and 0.9021 for a learned patch CNN. On brain LGG MRI the bank raises test ROC-AUC from 0.6903 to 0.7624, with 0.7335 retained after merging. A controlled sweep over the merged bank size shows that on breast data most of the improvement over prior single-kernel results comes from the multi-statistic readout rather than from kernel count, whereas on brain MRI kernel count is the dominant factor. These results indicate that a handful of interpretable radial filters, read out appropriately, retains substantial predictive power and provides a meaningful alternative or complement to black-box CNNs.

Future work should add repeated-seed confidence intervals and cross-validated merge selection, evaluate on prospective external cohorts, run the CNN reference on brain MRI to complete the comparison, quantify heatmap localization with Dice or free-response ROC, and examine whether the readout statistics themselves can be learned rather than fixed.

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